01 CPP FUNDAMENTALS — C++ Programs Reference PDF

Annotated C++ Source Programs — 01 CPP FUNDAMENTALS

Program: `01_first_classroom_program.cpp`

C++ SOURCE FILE • 01_first_classroom_program.cpp

```cpp
// Program Title: Standard Structure of a C++23 Classroom Program
// Topic: C++ Fundamentals
// Purpose: Demonstrate #include <iostream>, int main(), and console output.
// Target C++ Standard: C++23 (compatible with C++17/C++20)
// Related Theory File: ../01_THEORY/01_compilation_pipeline_and_program_structure.md
// Program Classification: Category A – Valid and well-defined program

#include <iostream>
using namespace std;

int main() {
    cout << "Welcome to B.Tech Mathematics and Computing C++23 Reference Library!" <<
endl;
    return 0;
}

/*
Expected Deterministic Output:
Welcome to B.Tech Mathematics and Computing C++23 Reference Library!
*/
```

Program: `02_basic_input_output_math.cpp`

C++ SOURCE FILE • 02_basic_input_output_math.cpp

```cpp
// Program Title: Basic Arithmetic and Stream Chaining
// Topic: C++ Fundamentals — I/O Streams
// Purpose: Demonstrate variable output and chained << insertion operators.
// Target C++ Standard: C++23 (compatible with C++17/C++20)
// Related Theory File: ../01_THEORY/02_iostream_and_namespace_std.md
// Program Classification: Category A — Valid and well-defined program

#include <iostream>
using namespace std;

int main() {
    int a = 15;
    int b = 25;
    int sum = a + b;

    cout << "a = " << a << ", b = " << b << ", sum = " << sum << endl;
    return 0;
}

/*
Expected Deterministic Output:
a = 15, b = 25, sum = 40
*/
```

Program: `03_void_main_compile_error.cpp`

C++ SOURCE FILE • 03_void_main_compile_error.cpp

```cpp
// INTENTIONAL COMPILE-TIME ERROR DEMONSTRATION
// Program Title: Non-Standard 'void main()' Rejection in Standard C++
// Topic: C++ Fundamentals — main() Return Type Rule
// Purpose: Show why 'void main()' is rejected by a conforming C++ compiler.
// Target C++ Standard: C++23
// Related Theory File: ../01_THEORY/01_compilation_pipeline_and_program_structure.md
// Program Classification: Category B — Intentional compilation-error demonstration

#include <iostream>
using namespace std;

void main() { // COMPILE ERROR: '::main' must return 'int'
    cout << "Hello" << endl;
}

/*
Why Compilation Fails:
ISO C++ [basic.start.main] requires the global main function to have a return type of 'int'.
Replace 'void main()' with 'int main()' as shown in 01_first_classroom_program.cpp.
*/
```